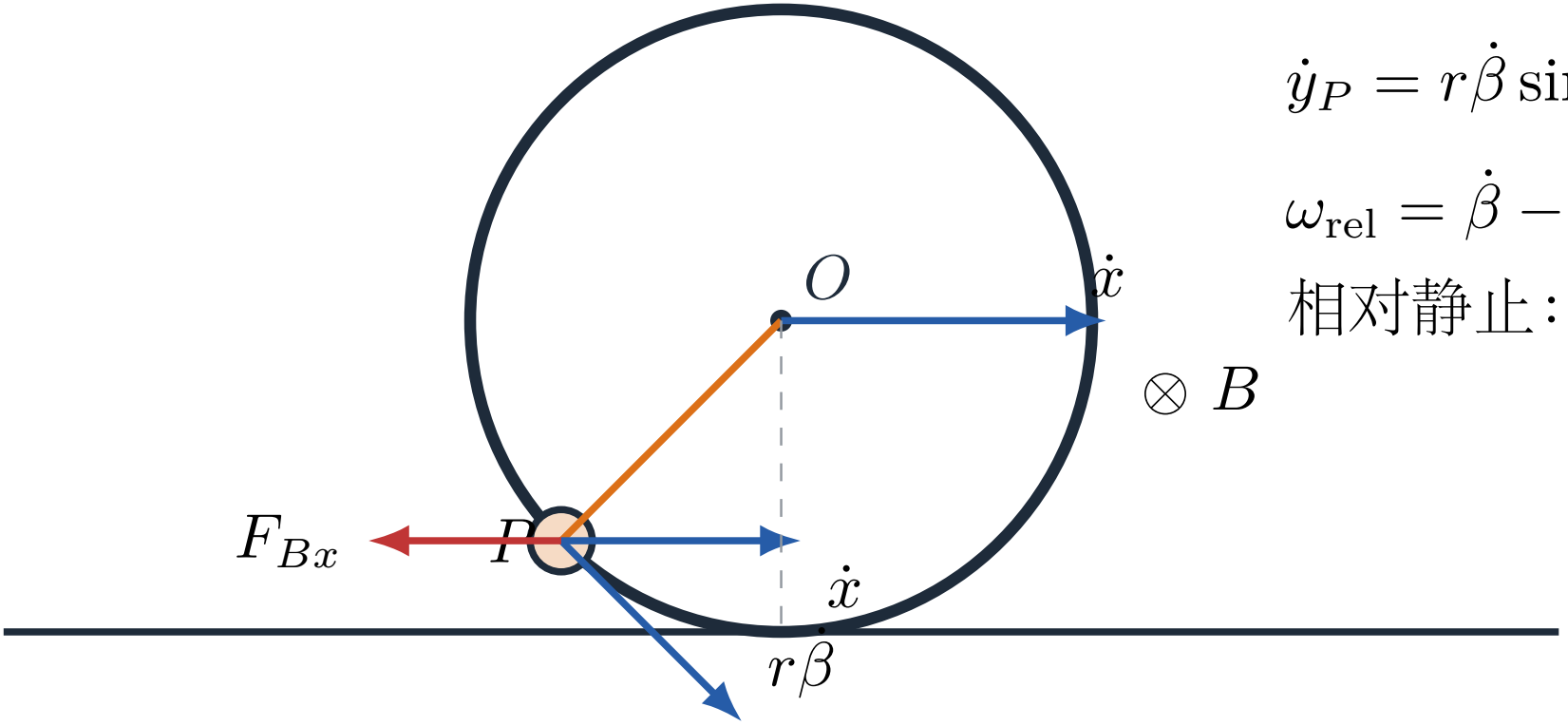
$$\dot{x}_P = \dot{x} - r\dot{\beta}\cos\beta$$

$$\dot{y}_P = r\dot{\beta}\sin\beta$$

$$\omega_{\text{rel}} = \dot{\beta} - \dot{x}/r$$

相对静止：$r\dot{\beta} = \dot{x}$



$$3\dot{x} - r\dot{\beta}\cos\beta = (qBr/m)\cos\beta$$